Rerouting Externalities: Flight Diversions and Local Health

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Who bears the local cost when a network reroutes activity?

Ordinary delay

- The flight lands at its scheduled airport.
- Residents experience a change in timing.

Flight diversion

- The flight lands at a different airport.
- Another community absorbs unplanned aircraft activity.

Does diversion-induced reallocation of aircraft activity increase emergency medical demand near receiving airports?

Key findings

1. **Diversions increase local EMS demand.**
   A one-standard-deviation increase in exposure raises total calls by 1.52% all day and 13.28% at night.

2. **Location and environmental exposure matter.**
   Effects are largest near airports. Delayed scheduled arrivals produce a different response. Wind and noise patterns suggest both physical channels.

3. **Capacity policy has consequences across airports.**
   EWR's 2016 slot relaxation was followed by more EWR-source diversions and higher EMS demand near JFK and LGA.

Contribution: the spatial incidence of network adjustment

Environmental spillovers

- Pollutants cross borders: Banzhaf and Chupp (2012), Fu et al. (2022), Jia and Ku (2019).
- Regulation relocates activity: Cai et al. (2016), Gibson (2019).
- Here, short-run operational responses relocate the activity itself.

Transportation networks and aviation externalities

- Congestion and propagation: Mayer and Sinai (2003).
- Local pollution and health: Schlenker and Walker (2016).
- Noise, health, and local welfare: Argys et al. (2020), Hener (2022), Allroggen et al. (2025), Kuminoff and Mandia (2025).

Network resilience can shift external costs to communities near substitute airports.

Outline

Introduction

Background

Identification

Health effects

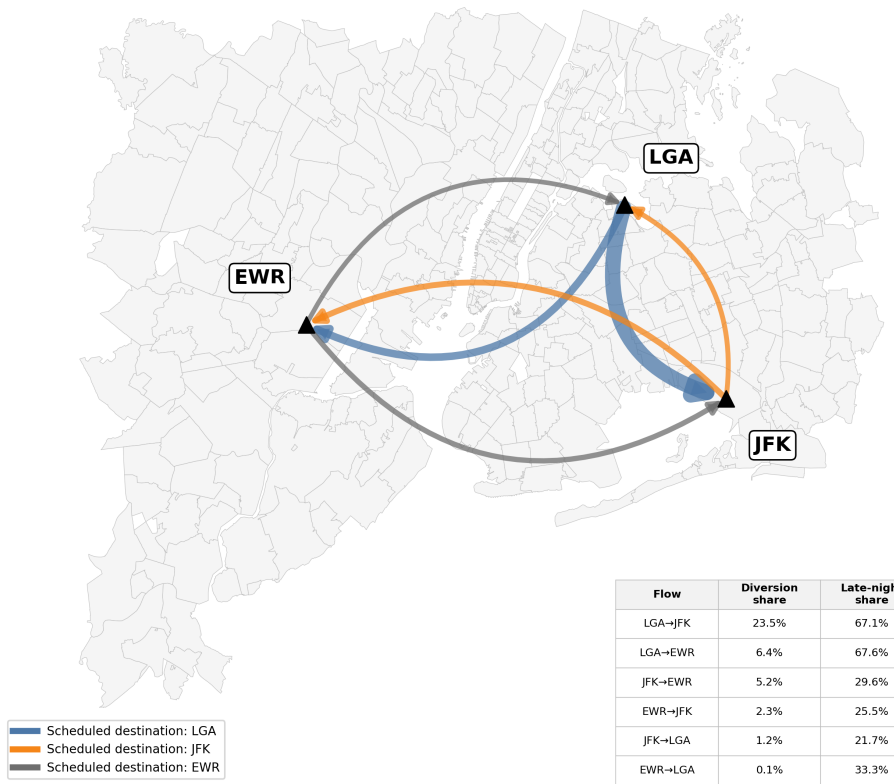
Capacity policy

Conclusion

Diversions repeatedly redistribute traffic within NYC

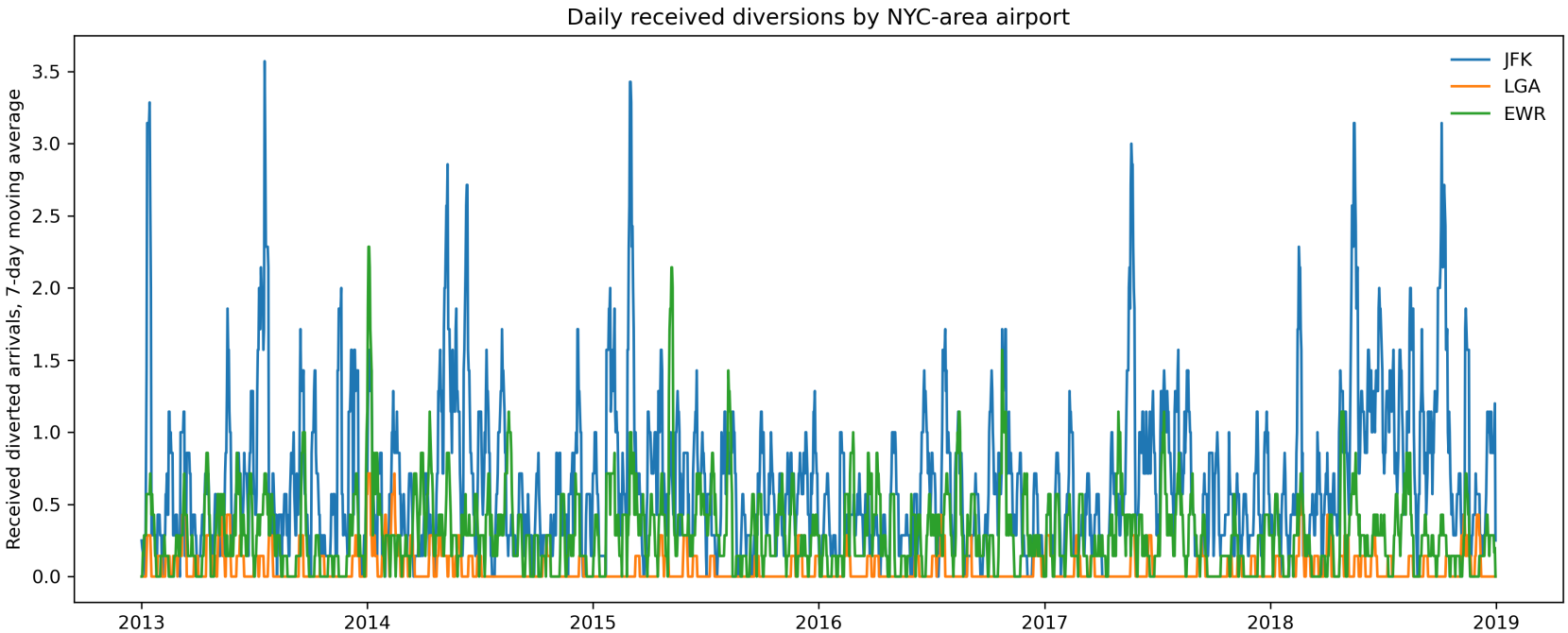
- At least one diversion is received in the system on **53% of days**.
- **68%** of received diversions were scheduled for another NYC-system airport.
- LGA-to-JFK is the largest internal flow, about **24%** of LGA diversions.

Pairwise Diversion Flows Among NYC-Area Airports, 2013-2018



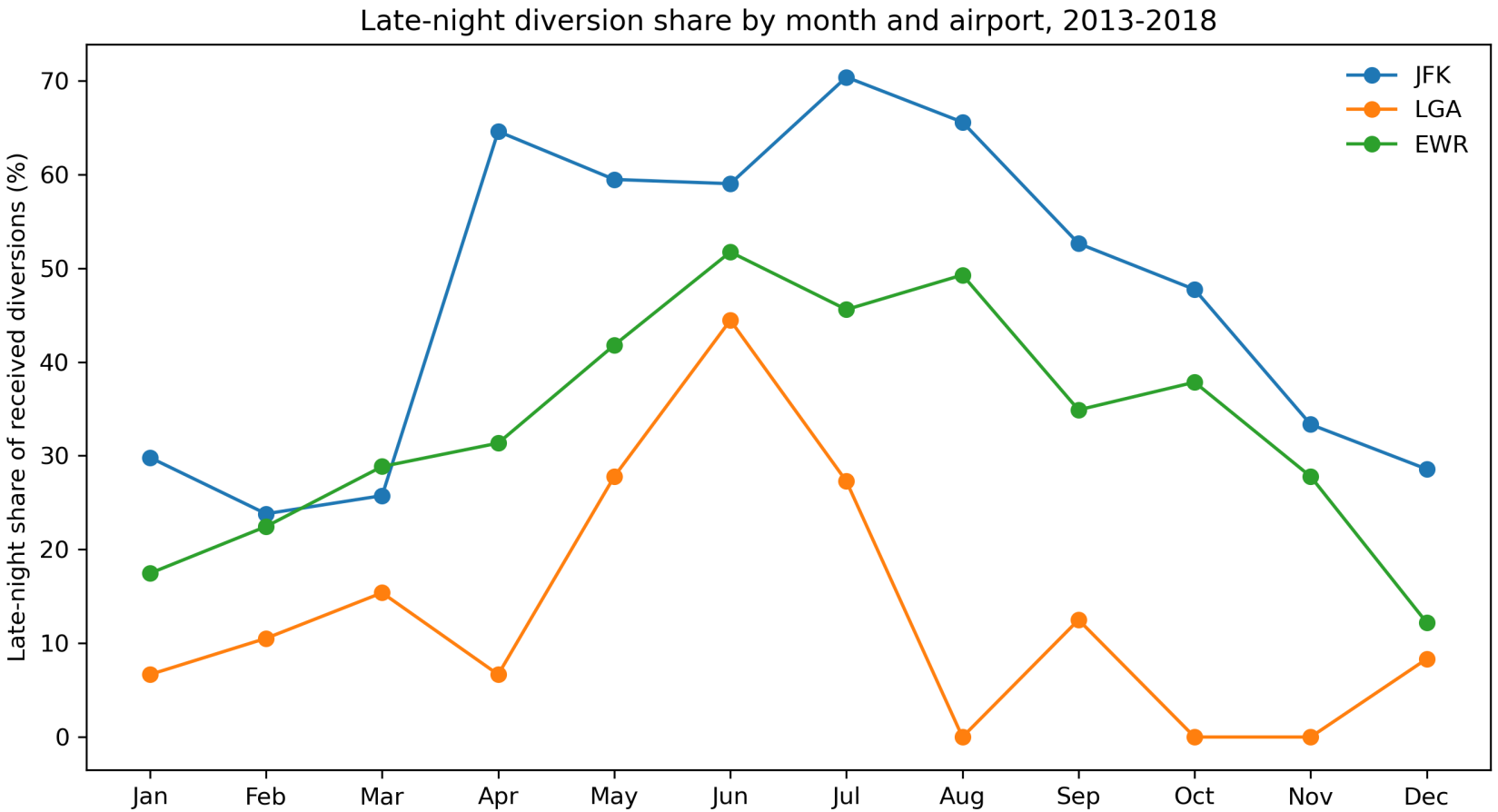
Flow	Diversion share	Late-night share
LGA→JFK	23.5%	67.1%
LGA→EWR	6.4%	67.6%
JFK→EWR	5.2%	29.6%
EWR→JFK	2.3%	25.5%
JFK→LGA	1.2%	21.7%
EWR→LGA	0.1%	33.3%

Diversion pressure varies sharply across days and airports



- Diversions arrive in spikes rather than as a steady traffic increment.
- About 67% of LGA-to-JFK and 68% of LGA-to-EWR diversions arrive late at night.

Late-night diversion share by month and airport, 2013–2018



Linking flight operations to neighborhood EMS demand

- **NYC EMS dispatches, 2013–2018**
  - ZCTA locations and incident times; total, mental health, and drug/alcohol calls.
  - NYC incident records: 8.56 million dispatches before exclusions; 449,155 ZCTA-days in the full panel.

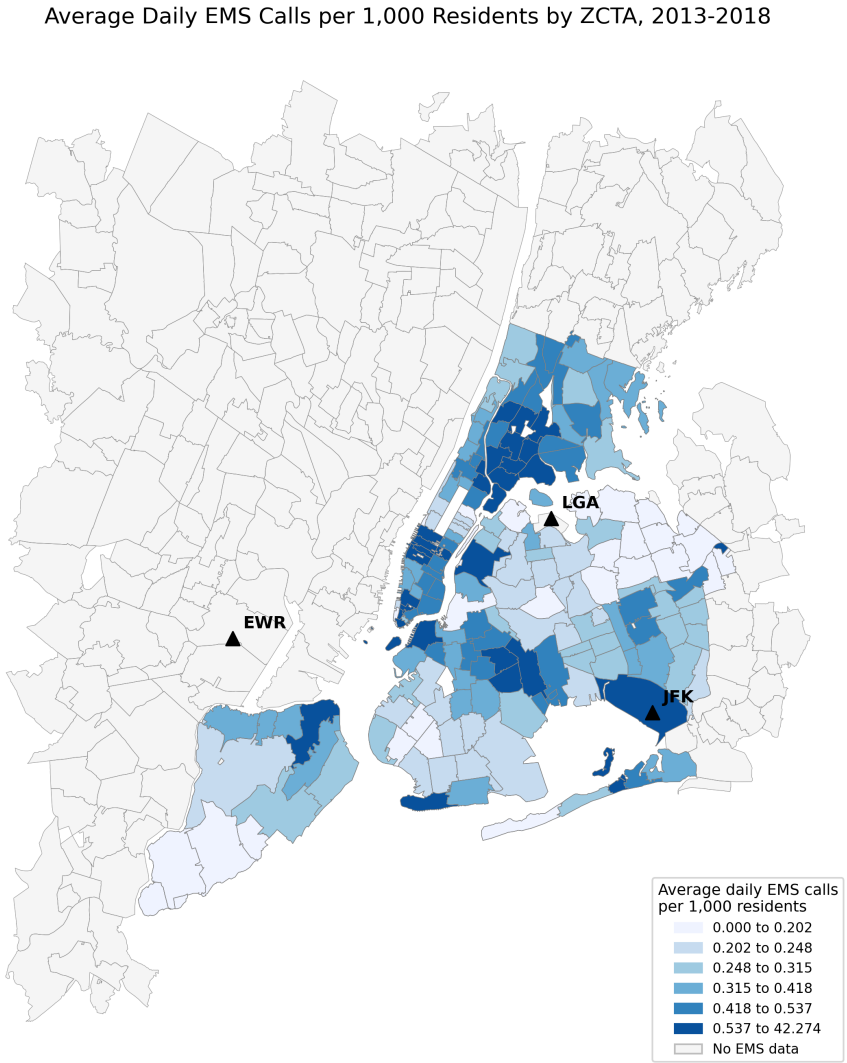
- **Flight operations and diversions**
  - BTS Airline On-Time Performance records, 2010–2018: flight dates, origins and destinations, scheduled and actual arrival times.
  - Cancellation/diversion indicators, arrival/departure delays, and diversion airports.
  - 2013–2018 records identify received diversions, airport operations, and unexpected late-night arrivals; 2010–2012 records supply network shares.

EMS calls measure acute demand for local response resources

Average ZCTA-day

- 15 total EMS calls.
- About 2 mental health and 1 drug/alcohol call.
- About 4 total calls at night.

Models use calls **per 1,000 residents**, retaining observations with zero calls.



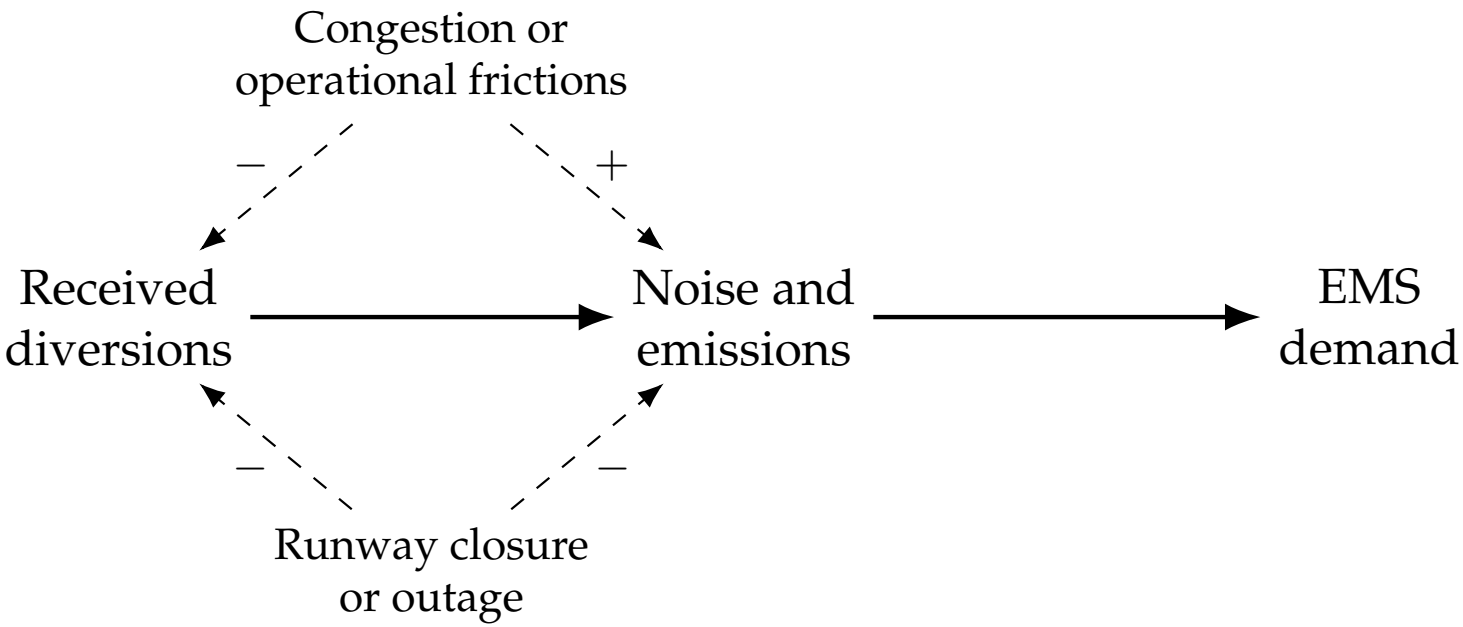
Neighborhood exposure combines activity at nearby airports

$$D_{zt} = \sum_{a \in \{JFK, LGA, EWR\}} w_{za} R_{at}$$

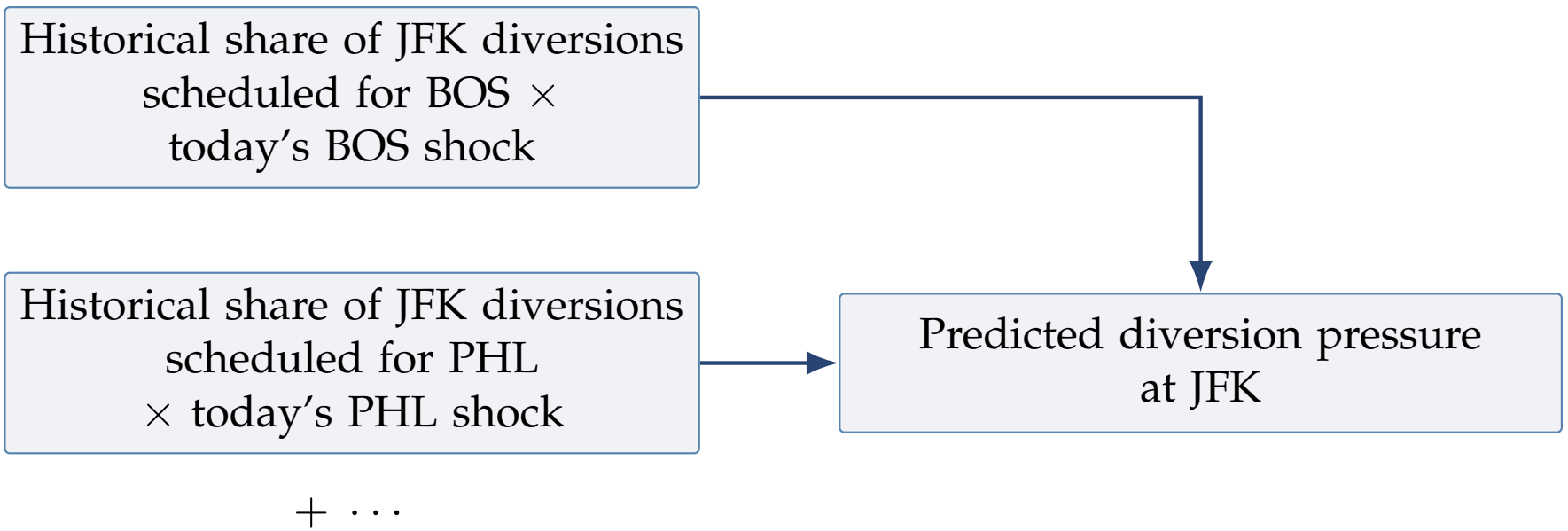
- R_{at}: diverted arrivals received by airport a on date t.
- w_{za}: predetermined inverse-squared-distance weights; zero beyond 30 km of the ZCTA centroid.
- Measure all-day activity and late-night activity (10 PM – 6 AM^{+1}), with EMS outcomes in the same window.
- Robustness: closest-airport exposure and a 10 km radius.

Why received diversions are endogenous

- Realized diversions are endogenous to local operating conditions.
- These shocks can produce bias in either direction.



Identification: Bartik shift-share instrument



- Actual diversions received in NYC never enter these source shocks.
- Fixed ZCTA–airport weights translate predicted pressure into local exposure.

Identification: Bartik shift-share instrument

$$Z_{at} = \sum_d \underbrace{s_{ad}}_{\substack{\text{2010–2012} \\ \text{diversion shares}}} \underbrace{D_{dt}^{(-NYC)}}_{\substack{\text{current destination} \\ \text{disruption}}} \qquad Z_{zt} = \sum_a w_{za} Z_{at}$$

- s_{ad}: share of diversions received by a that were scheduled for destination d in the pre-period.
- $D_{dt}^{(-NYC)}$: diversions from scheduled destination d, excluding flights received by **any** of JFK, LGA, or EWR.
- Late-night instruments use late-night shares and late-night destination shocks.

The instrument predicts received activity using disruption at historically connected scheduled destinations.

Control-function PPML

First stage

$$D_{zt} = \pi Z_{zt} + \theta S_{zt} + X'_{zt}\rho + \text{FE}_{zt} + v_{zt}$$

Second stage

$$\mathbb{E}[Y_{zt} \mid \cdot] = \exp\left(\beta^{CF} D_{zt} + \omega \widehat{v}_{zt} + \eta S_{zt} + X'_{zt}\gamma + \text{FE}_{zt}\right)$$

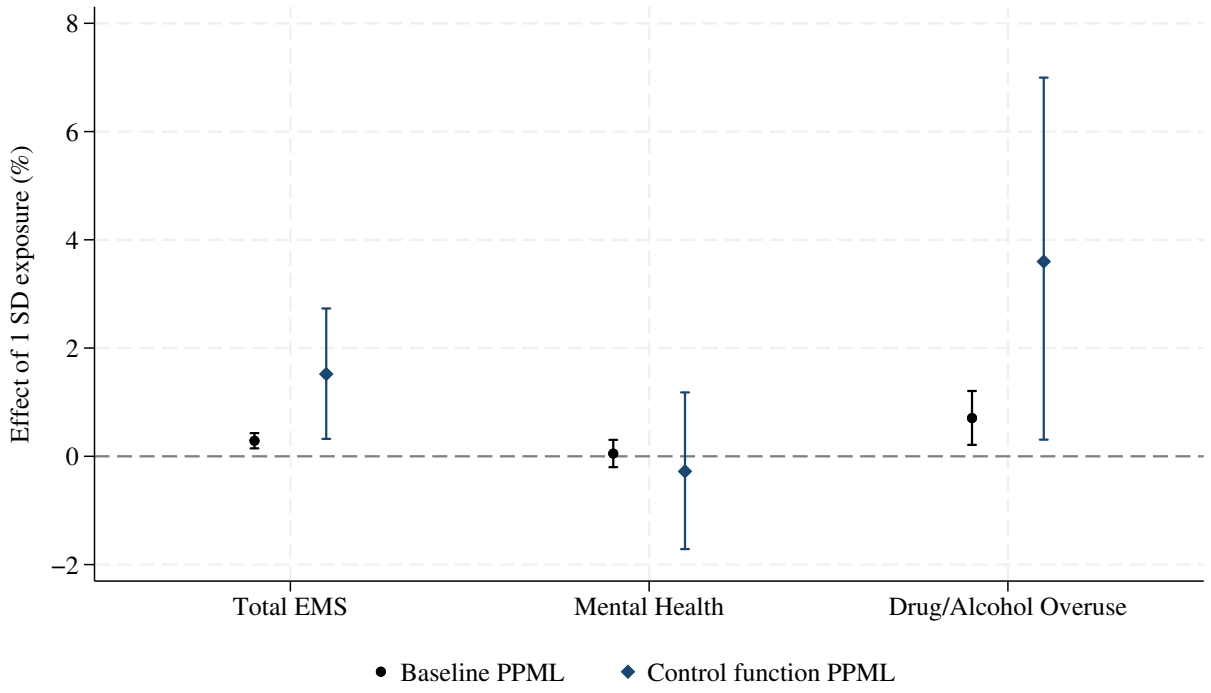
- Y_{zt} is EMS calls per 1,000 residents. S_{zt} is scheduled-arrival exposure.
- Weather: mean temperature and its square, minimum/maximum temperature, precipitation.
- Holidays, day of week, day of year, and ZCTA-by-year-month effects.
- Two-way clustered standard errors by ZCTA and date.

The instrument predicts diversion and delayed-arrival exposure

	All-day	Late-night	Unexpected late
Excluded instrument	0.1022***	0.1216***	0.2920***
	(0.0112)	(0.0264)	(0.0142)
First-stage F	83.27	21.22	422.85
Adjusted R^2	0.283	0.242	0.506
N	393,615	393,615	393,615

- First-stage F: excluded-instrument Wald statistic, the squared clustered t statistic.
- The common first-stage sample differs from some second-stage samples.

All-day diversions increase total and drug/alcohol EMS calls

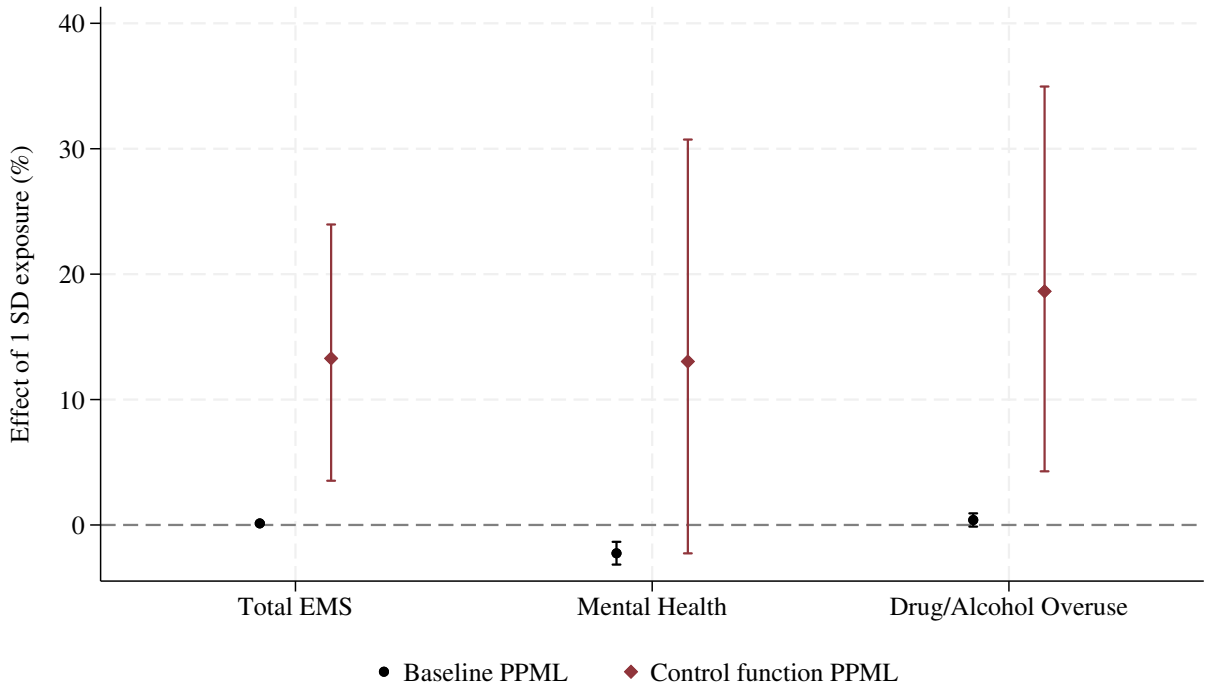


One-SD effects in CF-PPML: **1.52%** total EMS, −0.28% mental health, **3.60%** drug/alcohol overuse.

- One additional diversion landing implies about **22 extra calls across sample ZCTAs**, averaged over dates and receiving airports.

▸ Coefficients

The response is much larger for late-night diversions

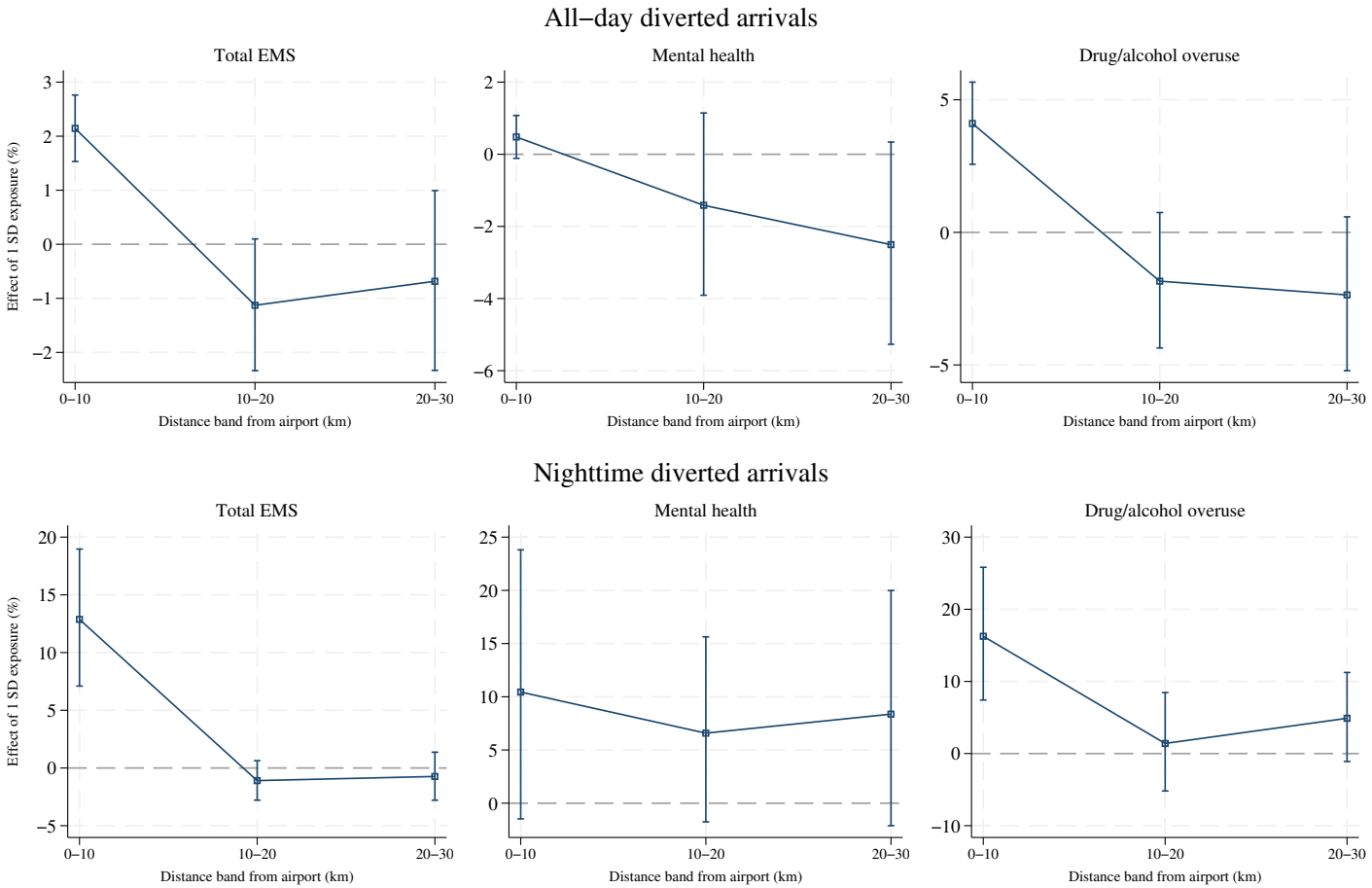


One-SD effects in CF-PPML: **13.28%** total EMS, **13.04%** mental health, **18.63%** drug/alcohol overuse.

- One additional late-night diversion landing implies about **78 extra calls across sample ZCTAs**, averaged over dates and receiving airports.

▸ Coefficients

Effects are concentrated near receiving airports



▸ Near-airport samples

Respiratory and cardiac effects: comparison of magnitudes

	Schlenker & Walker (2016)	All-day diversions	Late-night diversions
Exposure	One SD in taxi time	One SD in exposure	One SD in exposure
Outcome	Hospital admissions	EMS calls	EMS calls
Respiratory	About +1%; +6.91 admissions/day	+5.30%***	−0.49%
Cardiac	About +1%; +3.04 admissions/day	+2.42%*	+7.04%***

- **Schlenker and Walker (2016)**
  - Tarmac emissions at 12 California airports.
  - Different populations, outcomes, and exposure scales; the table provides a comparison of magnitudes.

▸ Supporting results

Delayed scheduled arrivals generate a different EMS response

Comparison: non-diverted flights scheduled before 10 p.m. but delayed into 10 p.m.–6 a.m., landing at their scheduled airport.

One-SD (%)	Total EMS	Mental health	Drug/alcohol
Late-night diversions	13.28***	13.04*	18.63***
Unexpected late arrivals	1.91	-1.88	8.37

- Instrument (following Schlenker and Walker, 2016): pre-period origin shares and severe departure-delay shocks, excluding NYC-bound flights.

The pattern supports a role for spatial reallocation beyond late arrival alone. The two exposures identify different margins.

▸ Delayed-arrival design

Some total EMS demand shifts forward from the next day

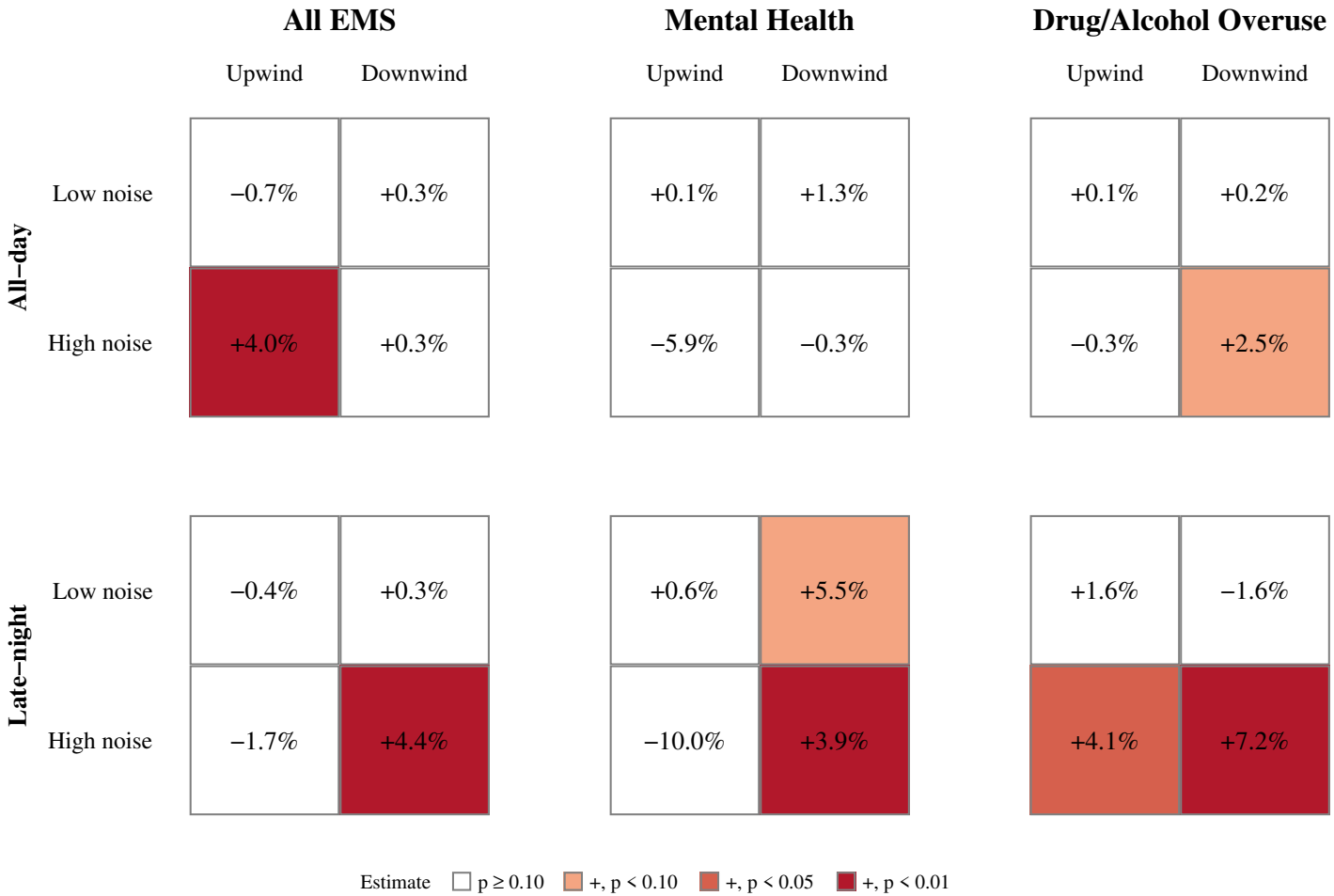
One-SD (%)	Total EMS	Mental health	Drug/alcohol
Exposed nighttime window	13.28***	13.04*	18.63***
Following daytime window	-5.96**	2.47	3.81

- Next-day outcomes cover 6 a.m.–10 p.m. after the exposed night.
- Total EMS falls, consistent with some temporal displacement.
- Mental health and drug/alcohol calls show no detectable offsetting decline.

Aircraft-noise contours around LaGuardia



Late-night patterns suggest both emissions and noise channels



Robustness checks and supporting evidence

- Near-airport samples ▸ Results
- Alternative weights and closest-airport assignment ▸ Results
- 10 km exposure cutoff ▸ Results
- Trim the top 1% of EMS outcomes ▸ Results
- Exclude small-population ZCTAs ▸ Results
- Shuffled outcomes and exposure ▸ Results
- Next-day displacement ▸ Results

Back-of-the-envelope cost calculation

One-SD all-day exposure shock

- 1.52% increase in total EMS.
- About **83 extra dispatches** when scaled to NYC's 2020 population.
- **$0.11–$0.19 million** in direct EMS response costs per shock day.

Annual costs

- About 33.5 high-exposure days per year in the sample.
- **$3.8–$6.4 million** in EMS response costs.
- **$5.2–$7.8 million** including expected ED care.

A service-resource calculation, using cost inputs from different years. It excludes broader health and welfare losses.

▸ Inputs and calculation

EWR's 2016 slot relaxation

- The FAA announced the change on April 6, 2016, effective October 29, 2016.
- EWR moved from Level 3 slot control to Level 2 schedule facilitation.
- JFK and LGA remained Level 3 slot-controlled airports.

	Before	After
EWR	Mandatory slot controls	Schedule facilitation
JFK and LGA	Mandatory slot controls	Mandatory slot controls

EWR experienced more delays after the policy transition

$$\mathbb{E}[Y_{at} \mid \cdot] = \exp(\tau Post_t \times \mathbf{1}\{a = EWR\} + \psi_a + \delta_t)$$

	Arrival delay 15+	Any arrival delay	NAS delay	Late-night arrivals	Late-night delays
Post × EWR	0.2072***	0.1721***	0.3808***	0.0161	0.1158***
	(0.0459)	(0.0642)	(0.0560)	(0.0289)	(0.0169)
Effect (%)	23.03	18.78	46.34	1.62	12.28

- EWR changes relative to JFK/LGA, with airport and date fixed effects.
- Little detectable change in late-night arrival volume itself.

EWR sent more diversions and received fewer

	EWR scheduled destination		EWR receiving airport	
	All day	Late night	All day	Late night
Post × EWR	0.1539***	0.3453***	-0.2872***	-0.6368***
	(0.0074)	(0.0220)	(0.1022)	(0.1460)
Effect (%)	16.63	41.24	-24.96	-47.10

- The table reports changes in diversion counts involving EWR after the policy transition.
- Percentage effects are calculated as $100[\exp(\widehat{\beta}) - 1]$.

Where did EWR-source diversions go?

EWR-source destination	All-day coefficient	Late-night coefficient
JFK	0.1141***	0.1773***
	(0.0071)	(0.0193)
LGA	1.2455***	—
	(0.0071)	
Outside NYC	0.1532***	0.3460***
	(0.0071)	(0.0193)

- JFK: +12.08% all day and +19.40% at night.
- LGA: +247.46% all day, from a pre-policy mean of 0.0007 flights per day; the late-night effect is not separately estimated.

Matched neighborhoods and the policy response

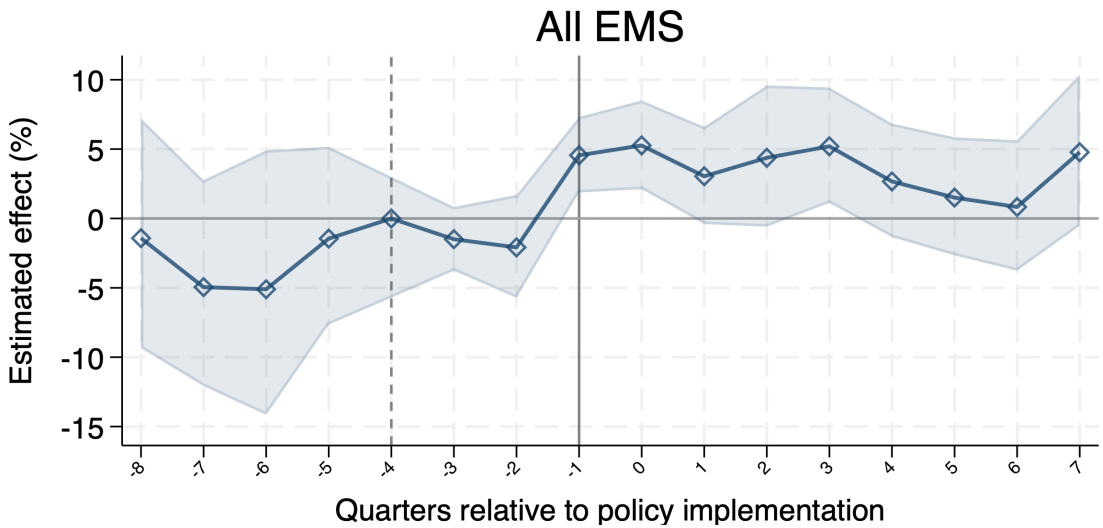
- **Treated:** within 15 km of JFK or LGA.
- **Controls:** 15–30 km from the corresponding airport. All neighborhoods more than 10 km from EWR.
- Match without replacement on pre-policy EMS levels and trends, exactly within the receiving-airport group.

$$\mathbb{E}[Y_{zt} \mid \cdot] = \exp\left(\sum_{q \neq 2016Q1} \kappa_q T_z \mathbf{1}\{t \in q\} + \alpha_z + \delta_{s(z),t}\right)$$

- Compare treated and matched controls within receiving-airport-by-date cells.
- Reference: 2016 Q1, the final full quarter before announcement.

▸ Matching balance

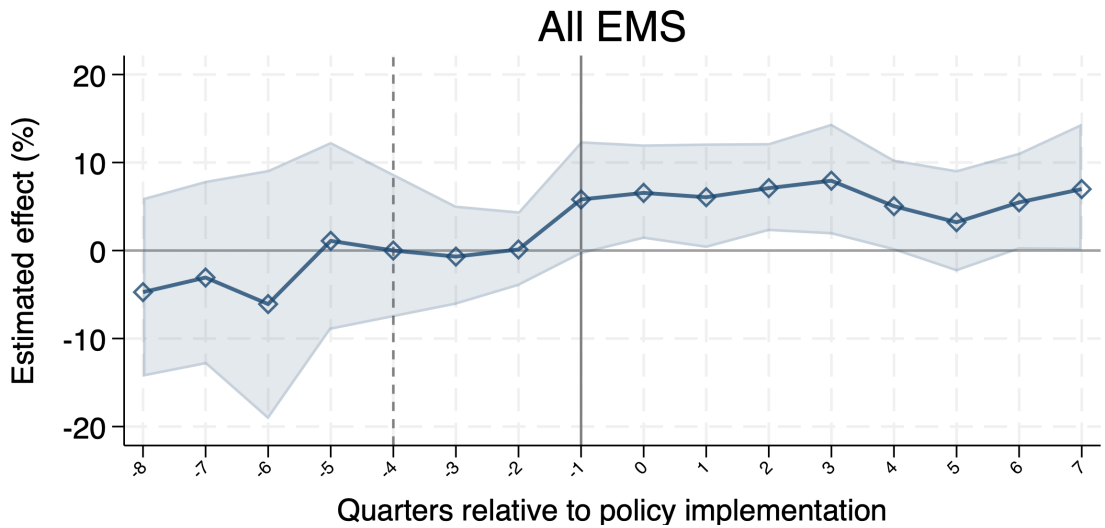
Total all-day EMS demand rises near the receiving airports



Post-implementation total EMS estimates are generally persistent, about **3–6%**. Mental health increases in the early post-policy quarters.

▸ All outcomes

Nighttime total EMS rises by roughly 5–8%

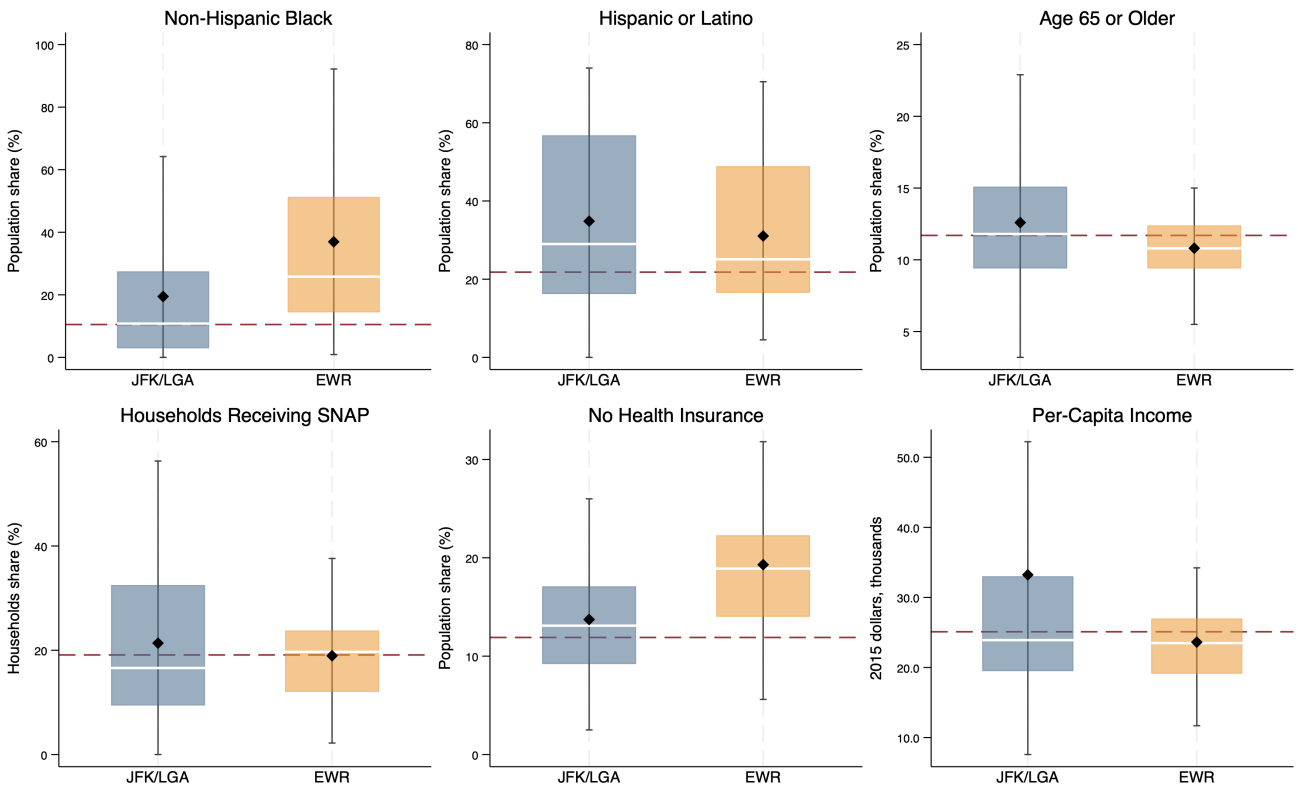


Nighttime mental health rises most near implementation. These are reduced-form policy responses that can include multiple operational changes.

▸ All outcomes ▸ EWR neighborhoods

Spatial reallocation and demographic burdens

- EWR: higher non-Hispanic Black and higher uninsured population shares.
- JFK/LGA: higher Hispanic population shares.
- Network routing transfers external costs across demographically distinct, predominantly minority communities.



Conclusion

1. **Received diversions increase local EMS demand.**
   Effects are especially large at night and close to receiving airports.

2. **Spatial reallocation helps explain the response.**
   Delayed scheduled arrivals differ. Wind and noise patterns suggest roles for both emissions and disturbance.

3. **A policy change can spread costs through the network.**
   EWR's slot relaxation was followed by more diversion activity elsewhere and higher EMS demand near JFK/LGA.

The relevant policy margin includes where disrupted flights land and which communities absorb the resulting activity.

All-day results: baseline and control-function estimates

	Total EMS	Mental health	Drug/alcohol
PPML coefficient	0.0056***	0.0010	0.0137***
	(0.0014)	(0.0025)	(0.0049)
One-SD (%)	0.29	0.05	0.71
CF coefficient	0.0294**	-0.0054	0.0687**
	(0.0118)	(0.0144)	(0.0320)
Residual coefficient	-0.0251**	0.0071	-0.0608**
	(0.0116)	(0.0172)	(0.0291)
One-SD (%)	1.52	-0.28	3.60
N	393,615	388,549	383,906

▸ Return

Late-night results: baseline and control-function estimates

	Total EMS	Mental health	Drug/alcohol
PPML coefficient	0.0040***	-0.0708***	0.0124
	(0.0015)	(0.0147)	(0.0084)
One-SD (%)	0.13	-2.25	0.40
CF coefficient	0.3895***	0.3807*	0.5339***
	(0.1435)	(0.2305)	(0.2056)
Residual coefficient	-0.3964***	-0.4605*	-0.5292**
	(0.1475)	(0.2473)	(0.2092)
One-SD (%)	13.28	13.04	18.63
N	392,347	374,330	371,883

▸ Return

Proximity checks: near-airport samples

	Total EMS	Mental health	Drug/alcohol
All day: within 10 km	0.0376***	0.0117**	0.0833***
	(0.0086)	(0.0050)	(0.0216)
One-SD (%)	2.06	0.64	4.66
All day: adjacent ZCTAs	0.0294***	0.0036**	0.0707***
	(0.0014)	(0.0016)	(0.0053)
One-SD (%)	2.74	0.33	6.72
Night: within 10 km	0.4572***	0.4982**	0.6427***
	(0.1138)	(0.2426)	(0.1689)
One-SD (%)	16.89	18.76	24.59
Night: adjacent ZCTAs	0.4903***	0.8138***	0.7218***
	(0.0297)	(0.0514)	(0.0301)
One-SD (%)	33.55	61.69	53.03

- Within 10 km: restrict ZCTAs by centroid proximity to a linked airport.
- Adjacent sample: airport-containing ZCTAs and their adjacent ZCTAs.
- Each cell uses the corresponding sample's exposure SD.

▸ Robustness ▸ Distance results

Alternative weights and primary-airport assignment

	Total EMS	Mental health	Drug/alcohol
All day: inverse distance	0.0263**	-0.0130	0.0663**
	(0.0118)	(0.0138)	(0.0316)
One-SD (%)	1.29	-0.63	3.29
All day: primary airport	0.0258**	0.0026	0.0573*
	(0.0116)	(0.0096)	(0.0301)
One-SD (%)	1.75	0.17	3.95
Night: inverse distance	0.3800**	0.3967*	0.5329**
	(0.1494)	(0.2327)	(0.2171)
One-SD (%)	12.27	12.88	17.60
Night: primary airport	0.3456**	0.3091	0.4761**
	(0.1457)	(0.2241)	(0.2134)
One-SD (%)	15.66	13.98	22.25

- Inverse-distance weights replace inverse-squared-distance weights.
- Primary-airport assignment uses an unweighted airport count.
- Remap diversion counts, the IV, and scheduled arrivals consistently.

▸ Return

A 10 km cutoff isolates more localized exposure

	Total EMS	Mental health	Drug/alcohol
All day: baseline	0.0294**	-0.0054	0.0687**
	(0.0118)	(0.0144)	(0.0320)
One-SD (%)	1.52	-0.28	3.60
All day: 10 km cutoff	0.0432***	0.0156***	0.0908***
	(0.0067)	(0.0039)	(0.0172)
One-SD (%)	2.32	0.83	4.98
Night: baseline	0.3895***	0.3807*	0.5339***
	(0.1435)	(0.2305)	(0.2056)
One-SD (%)	13.28	13.04	18.63
Night: 10 km cutoff	0.4846***	0.5037**	0.6612***
	(0.1066)	(0.2488)	(0.1656)
One-SD (%)	17.47	18.46	24.63

- Only airports within 10 km contribute to exposure, the IV, and scheduled arrivals.
- Restrict to ZCTAs with at least one airport inside this cutoff.
- This changes both exposure and sample, unlike a sample-only restriction.

▸ Return

Excluding the top 1% of EMS outcomes changes little

	Total EMS	Mental health	Drug/alcohol
All day: baseline	0.0294**	-0.0054	0.0687**
	(0.0118)	(0.0144)	(0.0320)
One-SD (%)	1.52	-0.28	3.60
All day: exclude top 1%	0.0288**	-0.0044	0.0683**
	(0.0113)	(0.0137)	(0.0293)
One-SD (%)	1.49	-0.23	3.59
Night: baseline	0.3895***	0.3807*	0.5339***
	(0.1435)	(0.2305)	(0.2056)
One-SD (%)	13.28	13.04	18.63
Night: exclude top 1%	0.3868***	0.4061*	0.5310***
	(0.1407)	(0.2387)	(0.1978)
One-SD (%)	13.20	13.98	18.55

- Remove observations above the outcome's 99th percentile and reestimate.
- The main findings do not depend on the largest observed EMS rates.

▸ Return

Excluding small-population ZCTAs gives similar estimates

	Total EMS	Mental health	Drug/alcohol
All day: baseline	0.0294**	-0.0054	0.0687**
	(0.0118)	(0.0144)	(0.0320)
One-SD (%)	1.52	-0.28	3.60
All day: population $\geq$ 100	0.0277**	0.0028	0.0666**
	(0.0122)	(0.0090)	(0.0300)
One-SD (%)	1.43	0.14	3.49
Night: baseline	0.3895***	0.3807*	0.5339***
	(0.1435)	(0.2305)	(0.2056)
One-SD (%)	13.28	13.04	18.63
Night: population $\geq$ 100	0.3865***	0.3738*	0.5259***
	(0.1405)	(0.2268)	(0.2024)
One-SD (%)	13.17	12.78	18.33

- Exclude ZCTAs with fewer than 100 residents.
- This addresses sensitivity to very small population denominators.

▸ Return

Placebo reassignments: all-day and late-night results

	Total EMS	Mental health	Drug/alcohol
All day: shuffled outcome	0.0131	-0.0198	0.1489
	(0.0495)	(0.0894)	(0.0912)
One-SD (%)	0.64	-0.96	7.56
All day: shuffled exposure	0.0020	0.0045	-0.0010
	(0.0014)	(0.0046)	(0.0092)
One-SD (%)	0.10	0.22	-0.05
Late night: shuffled outcome	-0.0224	-0.5016	-0.1953
	(0.2118)	(0.3465)	(0.3607)
One-SD (%)	-0.68	-14.13	-5.79
Late night: shuffled exposure	-0.0397	-0.0071	-0.0707
	(0.0278)	(0.0200)	(0.0453)
One-SD (%)	-1.20	-0.22	-2.13

The shuffled-outcome and shuffled-exposure coefficients are not statistically distinguishable from zero.

▸ Return

Next-day EMS after a late-night diversion shock

	Total EMS	Mental health	Drug/alcohol
CF coefficient	-0.1919**	0.0760	0.1164
	(0.0938)	(0.0528)	(0.0987)
Residual coefficient	0.1870**	-0.0886	-0.1130
	(0.0929)	(0.0569)	(0.0973)
Mean outcome	0.473	0.060	0.043
One-SD (%)	-5.96	2.47	3.81
N	393,350	386,786	379,084

Outcomes: calls per 1,000 from 6 a.m.–10 p.m. on $t+1$. Exposure: preceding 10 p.m.–6 a.m. First-stage $F = 20.44$.

▸ Return

Additional EMS categories: respiratory and cardiac calls

One-SD (%)	Respiratory	Cardiac
All-day diversions	5.30***	2.42*
Late-night diversions	-0.49	7.04***
Unexpected late arrivals	-1.65	-0.28

- All-day diversions increase respiratory EMS calls.
- Late-night diversions increase cardiac EMS calls.
- Unexpected late-arrival estimates are statistically imprecise.

Return

Unexpected late arrivals use origin-side disruption shocks

$$Z_{at}^{LNA} = \sum_{o} s_{ao}^{origin} \, Delay_{ot}^{(-NYC)}$$

- Severe departure delays: at least 60 minutes at origin o.
- Exclude NYC-bound flights from origin-side shock counts.
- Non-diverted arrivals delayed from before 10 p.m. into the night.

	Total EMS	Mental health	Drug/alcohol
CF coefficient	0.0021	-0.0021	0.0089
	(0.0019)	(0.0032)	(0.0061)
SD exposure	9.0530	9.0640	9.0688
One-SD (%)	1.91	-1.88	8.37

▸ Return

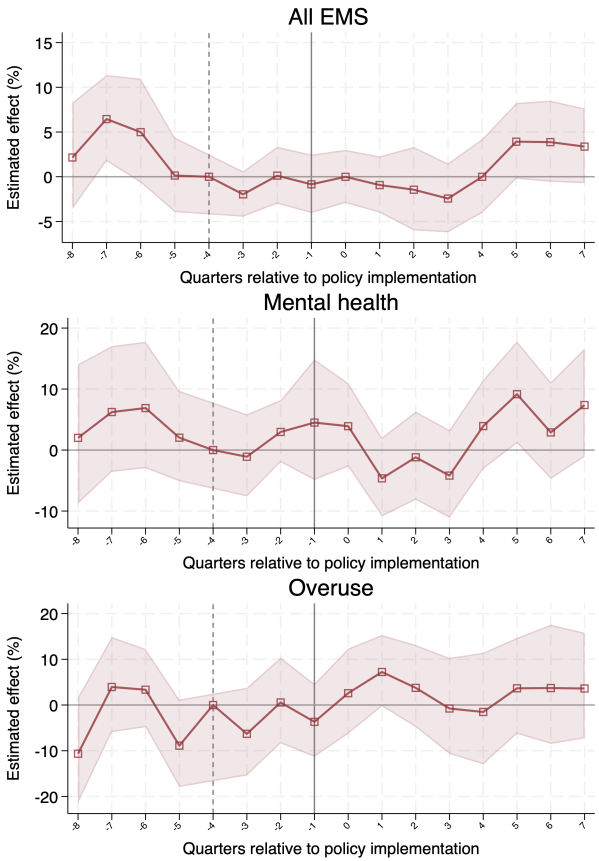
Matching balances pre-policy EMS levels and trends

Treated minus matched control	Difference	SE	p
Non-Hispanic Black share	0.354	0.107	0.006
Hispanic or Latino share	-0.033	0.037	0.389
Uninsured share	-0.012	0.011	0.276
Per-capita income ($1,000)	-0.107	7.998	0.990
All-day pre-policy EMS mean	0.075	0.164	0.658
All-day pre-policy EMS trend	-0.0020	0.0024	0.402
Late-night pre-policy EMS mean	0.018	0.038	0.640
Late-night pre-policy EMS trend	-0.0009	0.0008	0.297

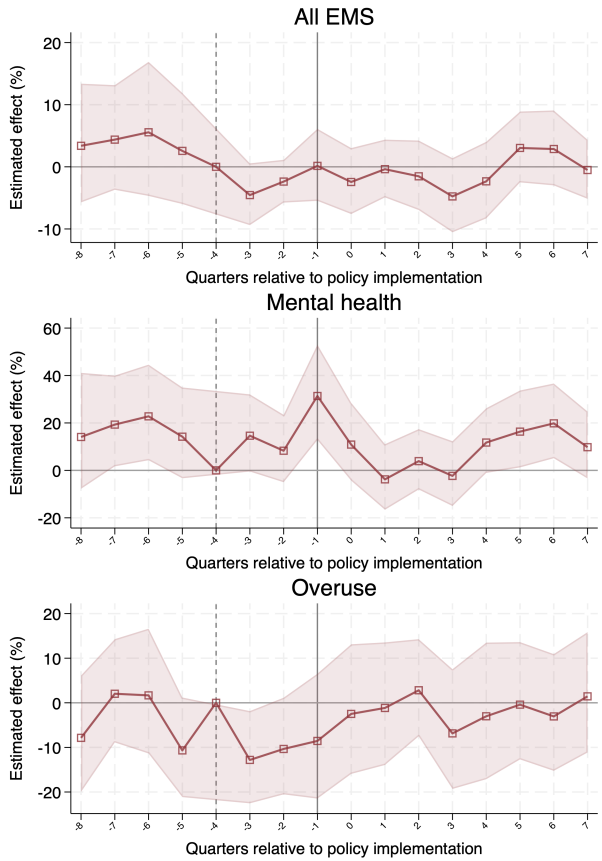
The non-Hispanic Black share remains higher in treated ZCTAs.

▸ Return

No clear offsetting EMS decline near EWR



All-day outcomes



Late-night outcomes

▸ Return

Cost calculation: inputs and annualization

Quantity	Benchmark
One-SD all-day effect on total EMS	1.52%
Mean EMS rate per 1,000	0.622
NYC population, 2020	8,804,190
Extra dispatches per shock day	83.24
EMS cost per response	\$1,369–\$2,280
Expected ED cost per incident	$0.62 \times \$820 \simeq \$508$
High-exposure days per year	33.5
Annual EMS / EMS plus ED	\$3.8–6.4m / \$5.2–7.8m

▸ Return

Model-implied effects of one additional diversion landing

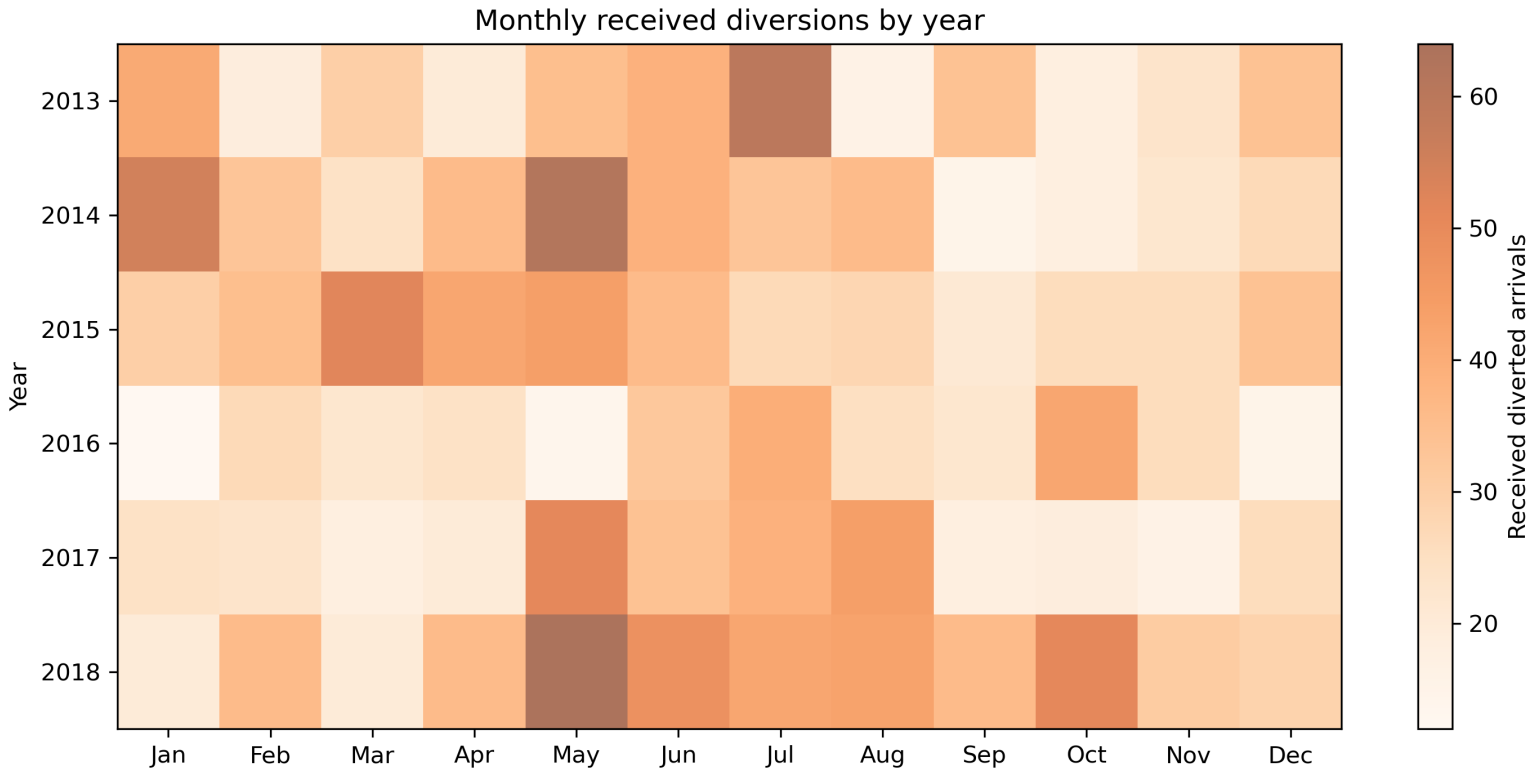
Model-implied extra calls	Total EMS	Mental health	Drug/alcohol
One all-day landing	22.35	-0.44	3.76
One late-night landing	78.09	7.21	11.15

$$\Delta_{at} = \sum_{z} \widehat{C}_{zt} \left[\exp(\widehat{\beta} w_{za}) - 1\right]$$

- Convert fitted rates into calls using the observed population.
- Average across dates and airports using diversion landing shares.
- Hold other regressors, including the CF residual, fixed.

▸ Return

Seasonality of received diversions



▸ Return

Selected references: spillovers, networks, and methods

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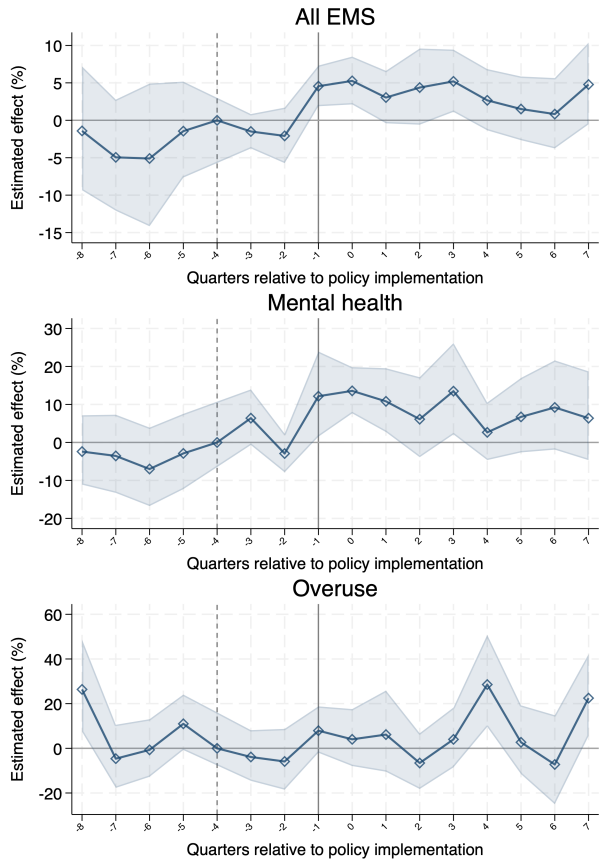
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Selected references: aviation, noise, and local outcomes

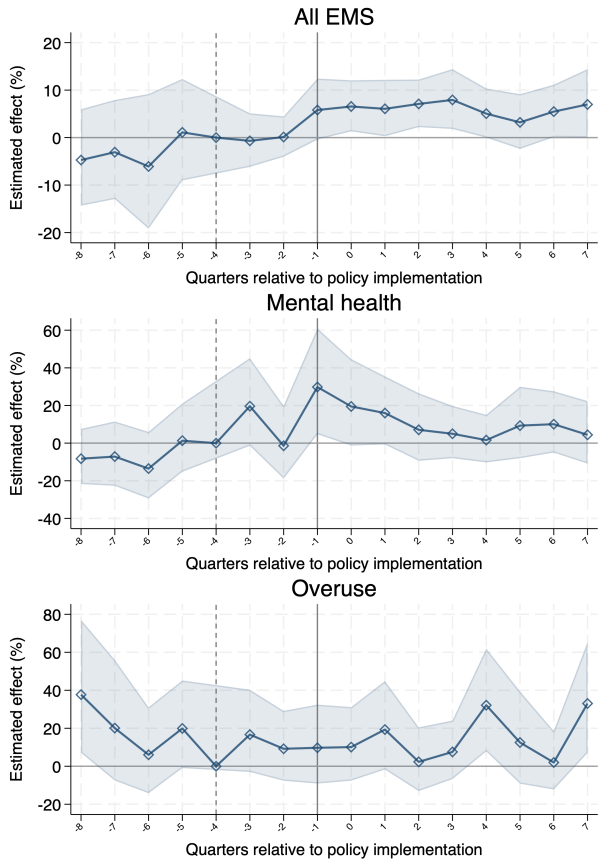
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▸ Return

JFK/LGA event study: all-day and late-night outcomes



All-day outcomes



Late-night outcomes

Percent effects with 90% confidence intervals. Reference period: 2016 Q1.

▸ All-day results ▸ Night results